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## Annex 1 — Process and Technology

**Purpose:** Describes the Urban Rig continuous superheated-steam pyrolysis process, its operating principles, and its key advantages over conventional waste treatment technologies.

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### 1.1 What the Urban Rig Does

The Urban Rig is a **continuous superheated-steam carbonisation device** that converts mixed solid waste into three marketable product streams — pyrolysis oil, biochar (carbide), and recovered metals — without combustion. The reactor operates at approximately 600°C in a sealed, oxygen-free environment, using superheated steam as the processing medium (Urban Rig Recycling Solutions ENG brochure, p.3; UR2000-JP technical specification, p.1).

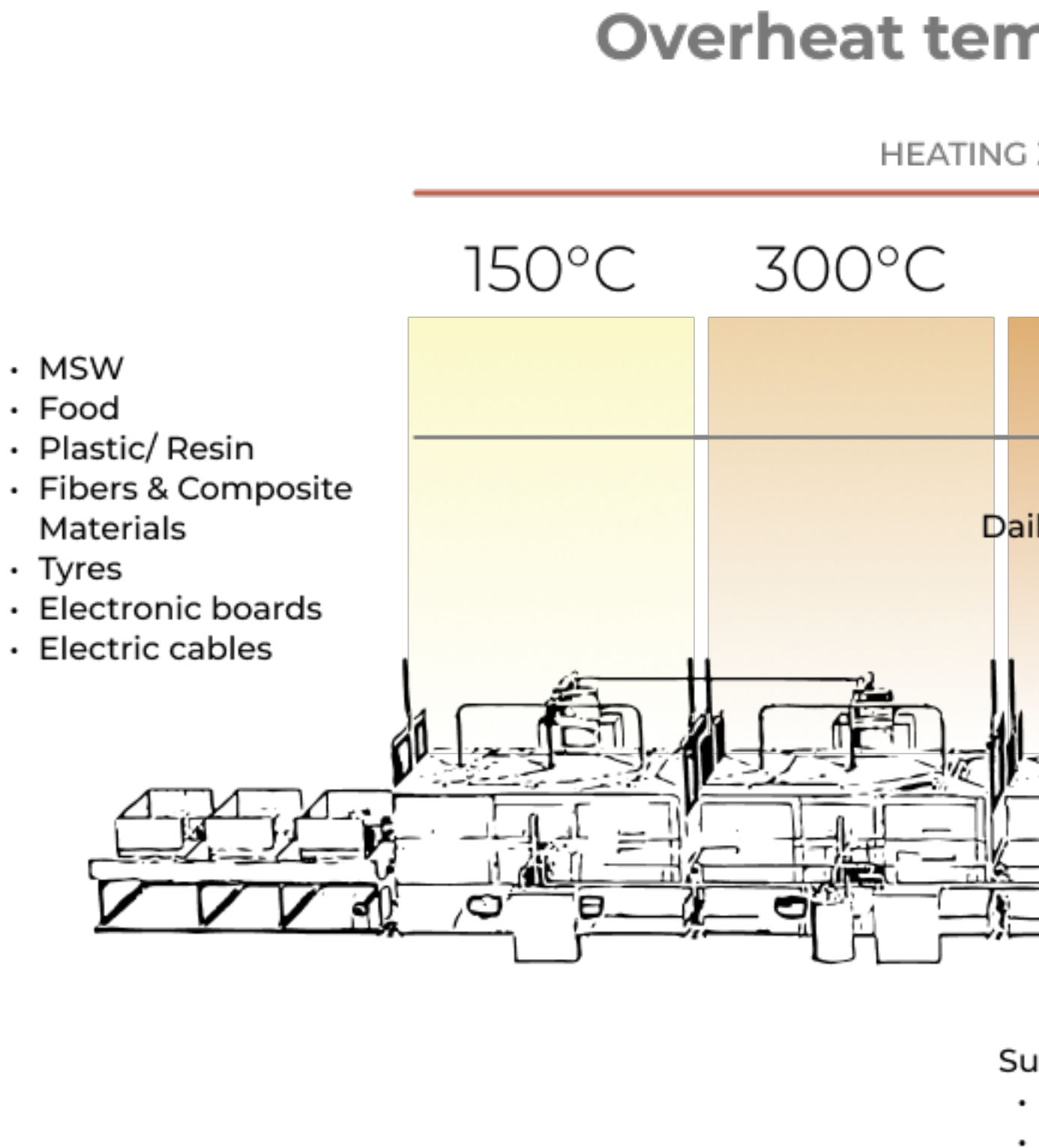
The process accepts unsorted, contaminated, and mixed waste — including plastics, food waste, paper, textiles, composites, and halogenated polymers such as PVC — without requiring pre-sorting or cleaning beyond basic size reduction and inert removal (Urban Rig Recycling Solutions ENG, p.6).

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## 1.2 How It Works

### 1.2.1 The 7-Zone Continuous Reactor



**Figure 1:** Illustration 2

The core of the system is a horizontal, multi-zone reactor through which waste is continuously conveyed on a chain-driven transport mechanism. As the waste moves from the inlet to the outlet, it passes through seven temperature zones of progressively increasing heat (Urban Rig Recycling Solutions ENG, pp.8–12; UR2000-JP, pp.2–4):

| Zone | Approximate Temperature | Primary Process                                                   |
|------|-------------------------|-------------------------------------------------------------------|
| 1    | ~100°C                  | Drying — moisture evaporates                                      |
| 2    | ~150°C                  | Light oil vapour collection (C5–C8 naphtha range)                 |
| 3    | ~250–300°C              | Gas oil / diesel oil vapour collection (C8–C16 middle distillate) |
| 4    | ~400–450°C              | Heavy oil vapour collection (C16+ waxes and paraffins)            |
| 5–6  | ~500–600°C              | Final carbonisation — solid residue becomes biochar/carbide       |
| 7    | ~600°C                  | Cooling zone — char exits reactor as solid granules               |

Oil vapours at each temperature point are drawn off through separate solenoid valves and condensed independently, producing three distinct oil grades rather than a single crude stream (Urban Rig Recycling Solutions ENG, p.15). This built-in fractional separation is a key design advantage (see section 1.4.3).

### 1.2.2 Superheated Steam Environment

The reactor is flooded with superheated steam at approximately 600°C. This serves three purposes:

1. **Oxygen exclusion.** Steam displaces all atmospheric oxygen inside the reactor, preventing combustion and the formation of dioxins, furans, and NO<sub>x</sub> — pollutants that require oxygen to form (Urban Rig Recycling Solutions ENG, p.14).
2. **Thermal cracking medium.** The steam provides uniform heat transfer to the waste, more efficient than radiative heating alone, ensuring consistent pyrolysis across the reactor cross-section (UR2000-JP, p.3).

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3. **Chemical activation.** Superheated steam promotes the cracking of long-chain polymers into shorter hydrocarbon chains, improving oil yield and quality compared to dry pyrolysis (Urban Rig Recycling Solutions ENG, p.14).

### 1.2.3 Off-Gas Recycling

Non-condensable gases generated during pyrolysis — primarily hydrogen, methane, ethane, ethylene, propane, propylene, and carbon monoxide (GC analysis report B2002600, 2020-12-05, collected at the Asada Shokai test facility, Noda, Japan) — are captured and recycled back to the reactor's burner system as supplementary fuel. This reduces external fuel consumption and contributes to the system's energy balance (Urban Rig Recycling Solutions ENG, p.12; energy balance test data from Asada Shokai facility, pp.35–36).

### 1.2.4 Continuous Operation

Unlike batch pyrolysis systems that must be loaded, heated, processed, cooled, and unloaded in discrete cycles, the Urban Rig operates continuously 24 hours per day. Waste enters at one end and products exit at the other in a steady flow. This continuous design delivers a throughput advantage of 3–14× over comparable batch systems at the same reactor volume (Urban Rig Recycling Solutions ENG, pp.5–6).

The URC-2000 reference model is rated at 200 tonnes per day volumetric capacity, equivalent to 140 tonnes per day at the standard waste density of 0.7 T/m<sup>3</sup> (UR2000-JP, p.4; Lufthansa worksheet assumptions, confirmed by user in Session 6).

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## 1.3 What Comes Out

The reactor produces four output streams from any mixed waste input:

### 1.3.1 Pyrolysis Oil (Four Fractions)

Oil vapours are collected at three temperature points and condensed separately, producing pre-fractionated liquid fuels (Urban Rig Recycling Solutions ENG, p.15):

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| Fraction               | Temperature | Carbon Range | Typical Share of Total Oil |
|------------------------|-------------|--------------|----------------------------|
| Light oil (kerosene)   | ~150°C      | C5–C8        | ~10%                       |
| Gas oil (diesel range) | ~300°C      | C8–C16       | ~80%                       |

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| Fraction                    | Temperature | Carbon Range | Typical Share of Total Oil |
|-----------------------------|-------------|--------------|----------------------------|
| Heavy oil (wax / paraffins) | ~450°C      | C16+         | ~10%                       |

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A fourth fraction — asphaltenes — is retained within the solid carbide residue and does not appear in the liquid output.

The dominant fraction (80%) is middle distillate in the diesel/kerosene range, confirmed by GC-MS analysis (Osaka Institute of Industrial Science and Technology) which identified dimethylheptane (C9), undecene (C11), and trimethylnonane (C12) as the principal compounds in the volatile fraction (Technical Findings Log §12).

Oil quality has been certified by NKKK (Nippon Kaiji Kentei Kyokai — Japan Marine Survey Association) at 43.27 MJ/kg calorific value, with sulfur at 0.014% and chlorine at 1 ppm (NKKK recycled fuel test report, JIS standards).

### 1.3.2 Biochar / Carbide

The solid residue exiting the reactor is a carbon-rich char. Its composition and calorific value depend on feedstock:

- **Plastic-derived char:** Dense, high-carbon, calorific value ~25 MJ/kg (UR2000-JP, p.3). One laboratory measurement on urethane-derived char returned 32.09 MJ/kg (Isumi Tech Co., Report R35-339439, JIS M 8814 bomb calorimeter).
- **Organic/cellulose-derived char:** Lower carbon density, estimated 15–25 MJ/kg depending on feedstock moisture and composition (pyrolysis literature: Demirbas 2004; IBI Biochar Standards). This range has not been validated on Urban Rig-produced MSW char.

The char is marketable as solid fuel (pelletised for cement kilns, coal replacement) or soil amendment (biochar for agriculture). Reactor heating is provided by purchased heating oil.

### 1.3.3 Recovered Metals

Metals in the waste stream — ferrous (steel), non-ferrous (aluminium, copper) — are recovered in **non-oxidised form** because the anoxic reactor environment prevents oxidation. This contrasts with incineration where metals are oxidised and must be recovered from toxic bottom ash at reduced quality and value (Urban Rig Recycling Solutions ENG, p.13; Eurostat waste incineration guidelines).

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### 1.3.4 Off-Gas

The non-condensable gas fraction is a combustible mixture of light hydrocarbons and hydrogen. Its calorific value has been measured at 5.54 MJ/kg (steam-diluted) across all feedstock types tested at the Asada Shokai facility (energy balance test data, pp.35–36; cross-validated against published pyrolysis literature — Bridgwater 2012, Chen et al. 2014). Off-gas is recycled as reactor fuel and is not a saleable output.

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## 1.4 Key Advantages

### 1.4.1 Accepts All Waste Types Without Pre-Sorting

The Urban Rig processes mixed, unsorted, dirty, laminated, and contaminated waste — including materials that no conventional recycling pathway can handle (Urban Rig Recycling Solutions ENG, pp.6–7):

- **PVC and halogenated plastics.** The reactor's catalytic layer adsorbs chlorine, and the superheated steam neutralises hydrochloric acid before it can damage downstream equipment or contaminate the oil output (Urban Rig Recycling Solutions ENG, p.14). Brominated flame retardants are similarly filtered (Asada Shokai test programme, 2022).
- **Contaminated food packaging.** Grease, food residues, and moisture are not a problem — the drying zone handles moisture, and organic contamination is simply converted to char.
- **Composite and multi-layer materials.** Tetra Pak cartons, metallised films, and other multi-material packaging that defeats mechanical recycling are fully processable.
- **Sanitary waste.** Diapers, sanitary pads, medical textiles — waste that is currently incinerated at high cost — can be processed.

### 1.4.2 Zero Dioxins, Near-Zero Emissions

Because the reactor operates in a sealed, oxygen-free environment, the chemical formation pathway for dioxins and furans is eliminated entirely — they cannot form without oxygen (Urban Rig Recycling Solutions ENG, p.14). This is fundamentally different from incineration, where dioxins are formed during combustion and must be captured by expensive filter systems.

Independent third-party exhaust gas analysis (KANSO Technos, report 20002610, 2020-12-05, samples collected 2020-11-25 at the Asada Shokai test facility) con-

firmed emissions 5–50× below French incineration standards across all measured pollutants:

| Pollutant       | Urban Rig Measured      | French Incineration Limit        | Factor Below Limit |
|-----------------|-------------------------|----------------------------------|--------------------|
| HCl             | <1 ppm                  | 10 mg/m <sup>3</sup> (~7 ppm)    | >7×                |
| SO <sub>x</sub> | <1 ppm                  | 50 mg/m <sup>3</sup>             | >50×               |
| NO <sub>x</sub> | 15 ppm                  | 200 mg/m <sup>3</sup> (~100 ppm) | >6×                |
| Particulates    | <0.002 g/m <sup>3</sup> | 10 mg/m <sup>3</sup>             | >5×                |

*Source: KANSO Technos Analysis Report 20002610; French incineration limits per Order of 20 September 2002 (Order of 20 September 2002).*

Odour testing (Total Environment System Co., Ltd., Osaka, 2020-11-27) measured an odour index of 14 at the emission source and <10 at property boundaries — within Japanese regulatory limits and compatible with proximity to residential or commercial areas.

### 1.4.3 Built-In Fractional Oil Separation

Unlike competing pyrolysis technologies that produce a single crude oil requiring downstream distillation, the Urban Rig’s 7-zone design collects oil at three distinct temperature points, delivering pre-fractionated products directly from the reactor (Urban Rig Recycling Solutions ENG, p.15). This provides three advantages:

1. **Higher realised prices.** Each fraction is sold at its grade-specific market price rather than at a crude pyrolysis oil discount (see Annex 3 for detailed pricing model).
2. **Reduced buyer processing costs.** Refineries and petrochemical operators can blend the appropriate fraction directly into their process stream without intermediate distillation.
3. **Multi-channel sales strategy.** Light oil targets naphtha cracker operators, gas oil targets diesel/kerosene blenders, and heavy oil serves marine fuel or hydrocracker feedstock markets — or can be burned on-site to power the reactor.

### 1.4.4 Microcomputer-Controlled Automation

The entire process is governed by a microcomputer automation system that monitors and adjusts temperatures, pressures, conveyor speeds, steam flow, and solenoid valve timing across all seven zones (Urban Rig Recycling Solutions ENG, p.17; UR2000-JP, p.5). This enables:



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- **Low staffing requirements.** Approximately 15 personnel operate a 200 T/day plant — significantly fewer than a comparable incineration facility (Urban Rig Recycling Solutions ENG, p.19).
  - **Consistent output quality.** Automated temperature control ensures repeatable oil fraction composition regardless of feedstock variability.
  - **Remote monitoring.** Operating parameters can be monitored and adjusted remotely, enabling centralised oversight of multiple units.

#### 1.4.5 Energy Requirements and Fuel Input

The Urban Rig system requires purchased heating oil for reactor heating. The system produces significant energy in its by-products (off-gas + char): at typical French MSW composition (23.7% plastic-like content), the combined off-gas and char energy output represents approximately 300% of the heating requirement (off-gas: ~24%, char: ~276%). While this surplus energy is not utilised for reactor heating, it represents significant potential energy value in the product streams, with all char sold on the market as solid fuel or agricultural biochar (see Annex 2 for detailed energy balance analysis).

#### 1.4.6 Modular and Scalable

Each Urban Rig unit is a self-contained, factory-manufactured module with a manufacturing lead time of approximately 10 months (Urban Rig Recycling Solutions ENG, p.19). Deployment does not require the multi-year construction timelines and billion-euro investments of a new incineration plant. Multiple units can be co-located on a single site to scale capacity incrementally, sharing building infrastructure and reducing per-unit CAPEX.

#### 1.4.7 Four Revenue Streams from a Single Input

Where incineration plants are pure cost centres — charging tipping fees and selling low-value electricity — the Urban Rig generates four distinct revenue streams from the same waste input:

1. **Gate fees** (waste intake charges)
2. **Fuel oil sales** (four fractions, sold at grade-specific prices)
3. **Char / biochar sales** (all char sold as fuel pellets or agricultural biochar)
4. **Recovered metals** (non-oxidised steel, aluminium, copper)

A fifth revenue stream — **carbon credits** — is available in jurisdictions with carbon pricing mechanisms (EU ETS, voluntary markets), based on CO<sub>2</sub> avoided versus the incineration baseline.

## 1.5 How It Compares

The following table provides a systematic comparison across the three dominant waste treatment pathways. The Urban Rig operates on a fundamentally different principle: converting waste into tradeable commodities rather than simply destroying it or generating low-grade heat.

| Factor                           | Landfilling                                               | Incineration (WTE)                                                 | Urban Rig (WTV)                                     |
|----------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|
| GHG emissions                    | High (methane, 80× CO <sub>2</sub> potency over 20 years) | Moderate-high (CO <sub>2</sub> , trace dioxins)                    | Low (biogenic CO <sub>2</sub> only, no methane)     |
| Energy recovery                  | None                                                      | Medium (electricity + district heat)                               | High (liquid fuels + solid fuel char)               |
| Residue                          | 100% remains as waste                                     | 20–30% toxic bottom ash + fly ash                                  | Minimal inert char (<5% of input)                   |
| Metal recovery                   | Lost or corroded                                          | Oxidised (recovered from ash at low quality)                       | Non-oxidised (steel, aluminium, copper)             |
| Feedstock requirement            | Low                                                       | Moderate (pre-treatment often needed)                              | Low — accepts dirty, mixed, composite               |
| Pollution control                | Leachate management, methane capture                      | Complex (filters, scrubbers, DeNO <sub>x</sub> , activated carbon) | Minimal (closed-loop anoxic process)                |
| Revenue model                    | Pure cost                                                 | Electricity sales (low margin)                                     | Four revenue streams (oil, char, metals, gate fees) |
| Dioxin formation                 | Not applicable                                            | Yes — requires post-formation capture                              | Impossible (zero oxygen)                            |
| Typical CAPEX (200 T/day equiv.) | €20–50M (land + civil works)                              | €300M–1B+ (Ivry-Paris XIII: €1.3B)                                 | ~€61M (turnkey including building)                  |

*Sources: Eurostat waste management guidelines; ADEME comparative studies; Urban Rig technical documentation; SYCTOM Ivry-Paris XIII project budget.*

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## 1.6 References

1. **Urban Rig Recycling Solutions ENG** — OneWorld Corporation product brochure (English edition). Cited for: reactor design (pp. 3, 5–8, 12, 14–15, 17, 19), fractional oil separation, emission-free environment, automation, staffing, scalability, and halogen filtration.
2. **UR2000-JP** — URC-2000 technical specification sheet (Japanese edition). Cited for: throughput (p. 4), reactor zones (pp. 2–4), automation (p. 5).
3. **GC analysis report B2002600** (2020-12-05) — Gas chromatography analysis of non-condensable off-gas, collected at the Asada Shokai test facility, Noda, Japan. Cited for: off-gas composition (§1.2.3).
4. **Asada Shokai test programme** (Noda, 2022) — 74-page test report with independent measurement by Land Brain Co., Ltd. Cited for: off-gas energy balance (pp. 35–36), halogen filtration validation.
5. **KANSO Technos Analysis Report 20002610** (2020-12-05, samples 2020-11-25) — Independent third-party exhaust gas analysis at the Asada Shokai test facility. Cited for: emission measurements (§1.4.2).
6. **Order of 20 September 2002 (Order of 20 September 2002)** — French regulation on incineration emission limits. Cited for: comparison benchmarks (§1.4.2).
7. **Total Environment System Co., Ltd.** (Osaka, 2020-11-27) — Odour testing at the Asada Shokai facility. Cited for: odour index (§1.4.2).
8. **NKKK (Nippon Kaiji Kentei Kyokai)** — Recycled fuel test report, JIS standards. Cited for: oil quality certification (§1.3.1).
9. **Osaka Institute of Industrial Science and Technology** — GC-MS analysis of volatile oil fraction. Cited for: oil compound identification (§1.3.1).
10. **Isumi Tech Co., Report R35-339439** — Bomb calorimetry (JIS M 8814) of urethane-derived char. Cited for: char CV 32.09 MJ/kg (§1.3.2).
11. **Demirbas, A.** (2004) — Pyrolysis literature on cellulose-derived char CV. Cited for: organic char range 15–25 MJ/kg (§1.3.2).
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13. **Bridgwater, A.V.** (2012) — “Review of fast pyrolysis of biomass and product upgrading,” *Biomass and Bioenergy*. Cited for: off-gas cross-validation (§1.3.4).
14. **Chen, D. et al.** (2014) — “Pyrolysis technologies for municipal solid waste,” *Waste Management*. Cited for: off-gas cross-validation (§1.3.4).
15. **Eurostat** — Waste management guidelines and incineration benchmarks. Cited for: metal oxidation comparison (§1.3.3), waste treatment comparison table (§1.5).
16. **ADEME** — Comparative waste treatment studies. Cited for: technology comparison (§1.5).
17. **SYCTOM** — Ivry-Paris XIII replacement project budget (€1.3B). Cited for: CAPEX

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comparison (§1.5).

18. **Lufthansa worksheet** — Operational assumptions confirmed in Session 6.  
Cited for: effective throughput 140 T/day at 0.7 T/m<sup>3</sup> density (§1.2.4).
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**End of Annex 1**